

COURSE SYLLABUS

CSC12002 – Advanced Database Systems

1. GENERAL INFORMATION

Course name:	Advanced Database Systems
Course name (in Vietnamese):	Cơ sở dữ liệu nâng cao
Course ID:	CSC12002
Knowledge block:	Fundamentals of Databases
Number of credits:	4
Credit hours for theory:	45
Credit hours for practice:	30
Credit hours for self-study:	90
Prerequisite:	
Prior-course:	Introduction to Databases (Cơ sở dữ liệu)
Instructors:	

2. COURSE DESCRIPTION

This course aims to equip students with advanced knowledge in the field of relational database design. The content of the course focuses on four phases in database design: collecting and analyzing user requirements, conceptual database design, logical database design, and physical database design. In conceptual database design, students learn how to determine user requirements for data and develop data models to represent those requirements. Logical specifications, which map the conceptual requirements into the data model associated with a specific database management system. In the conceptual and logical design phases, the role of functional dependency will be discussed, especially related to the issues of integrity constraints checking, quality assessment of database schema and selecting the right database schema associated with information system requirements. In physical database design,

students learn how to transform the requirements for data storage developed during database analysis into specifications to guide database implementation. Physical specifications, which indicate all the parameters for data storage that are then used as input for database implementation. During this phase, the content of course also focuses on how to select the appropriate physical techniques (index, data fragment, partition) for efficient queries on a specific relational database management system such as MS SQL, Oracle. In addition, the content of the course also introduces some new popular databases today, such as NoSQL database, geographic information database and other types of databases on the Internet.

3. COURSE GOALS

At the end of the course, students are able to:

ID	Description	Program LOs
G1	Practice teamwork skills.	2.2.1; 2.2.2
G2	Practice reading English documents related to relational database design.	2.4.3; 2.4.5
G3	Practice critical and creative thinking in relational database design.	2.1.4; 2.1.5; 4.1.1; 4.1.2; 4.1.3; 4.1.4
G4	Explain the meaning and importance of relational databases design.	1.3.5
G5	Practice and apply the basic knowledge for the relational database design phases such as data models, functional dependency, normal form, ...	1.3.5
G6	Apply some techniques to get user requirements.	4.2.2; 5.1.1
G7	Execute the tasks in the conceptual and logical database design.	2.3.1, 2.3.2; 4.1.2; 5.2.1; 5.2.2; 1.4.10; 1.4.11; 1.4.14
G8	Execute the tasks in the physical design phases.	1.2.1; 1.3.1; 5.2.1; 5.2.2; 5.3.1; 5.3.2; 6.1.2; 1.4.11; 1.4.14; 1.4.10
G9	List some modern popular databases currently.	1.3.5; 2.1.8; 1.4.10; 1.4.14

4. COURSE OUTCOMES

CO	Description	ITU
G1.1	Practice skills: creating group, group management, negotiation and conflict resolution skills in group discussion.	U
G1.2	Practice writing the required technical documents.	U
G1.3	Practice presenting the required results of relational database design phases.	U
G2.1	Use key terms in English in database design phases.	TU
G2.2	Interpret documents (slide, book chapter) in English.	TU
G3.1	Apply creative thinking in solving problems in relational database design process.	TU
G3.2	Apply critical thinking (comparison and contrast, deduction, evaluation) in the relational database design process.	U
G4.1	Describe the importance of relational database design.	I
G4.2	Describe the Database Life Cycle.	I
G4.3	Describe business rules are and how they influence database design.	I
G5.1	Demonstrate data models: entity relationship model (ERM), relational data model, and how to convert from ERM to relational data model.	TU

CO	Description	ITU
G5.2	Define the concept of functional dependencies (FDs) and know how to use FDs to declare data constraints.	TU
G5.3	Recognize data duplication on the schema and the anomalies caused by data duplication.	TU
G6.1	List the techniques used to get user requirements.	T
G6.2	Practice getting user requirements.	T
G7.1	Practice designing conceptual data schema using Entity Relationship Model.	TU
G7.2	Practice converting ER diagram to Relational Data Model.	TU
G7.3	Describe data dependency by using functional dependencies.	TU
G7.4	List the normal forms and practice transforming from lower normal forms to higher normal forms (normalization).	TU
G7.5	Use normalization to decompose a relation with anomalies into well-structured relations.	TU
G8.1	Concisely define each of the key terms using physical level: field, data type, denormalization, horizontal partitioning, vertical partitioning, physical file, tablespace, ...	TU
G8.2	Describe the physical database design process, its objectives, and its deliverables.	TU
G8.3	Choose storage formats for attributes from a logical data model and select an appropriate file organization by balancing various important design factors.	TU

CO	Description	ITU
G8.4	Describe three important types of file organization and Describe the purpose of indexes and the important considerations in selecting attributes to be indexed.	TU
G8.5	Translate a relational data model into efficient database structures, including knowing when and how to denormalize the logical data model.	TU
G8.6	Construct physical data schema into DBMS (MS SQL) to check performance.	TU
G9.1	List some modern databases such as real-time database, NoSQL, Blockchain, XML, HDT database, geographic database, ...	I

5. TEACHING PLAN THEORY

ID	Topic	COs	Teaching/Learning Activities
1	Overview of Relational Database Design	G2, G4.1, G4.2, G4.3	Lecturing
2	Data Models: Entity Relationship Model	G5.1, G2	Lecturing Q&A
3	Data models: Enhanced Entity Relationship Model Relational Data Model	G5.1, G2	Lecturing Demonstration, discussion
4	Conceptual Database Design (part 1)	G6.1, G6.2, G7.1, G1, G2, G3	Lecturing Case study Demonstration, discussion

ID	Topic	COs	Teaching/Learning Activities
5	Conceptual Database Design (part 2)	G6.1, G6.2, G7.1, G1, G2, G3	Lecturing Case study, Demonstration, discussion
6	Data Dependency & Normal Form	G5.2, G5.3, G1, G2, G3	Lecturing Question & answer Discussion
7	Logic Database Design	G7.2, G7.3, G7.4, G7.5, G1, G2, G3	Lecturing Case study, Demonstration, discussion
8	Physical Database Design (part 1)	G8.1, G8.2, G8.3, G1, G2, G3	Lecturing Q&A, discussion
9	Physical Database Design (part 2)	G8.4, G8.5, G8.6, G1, G2, G3	Lecturing Q&A, discussion
10	Advanced Topics	G9.1, G2	Lecturing Demonstration
11	Review		Q&A, Discussion Project submitted

LABORATORY

For the practical laboratory work, there are 10 weeks which cover similar topics as it goes in the theory class. Each week, teaching assistants will explain and demonstrate key ideas on the corresponding topic and ask students to do their lab exercises either on computer in the lab or at home. All the lab work submitted will be graded. There would be a final exam for lab work.

6. ASSESSMENTS

ID	Topic	Description	COs	Ratio (%)
A1	Assignments			20%
A11	Quizzes and exercises	Small quizzes and/or exercises in class for each topic.		15%
A12	Homeworks	Studying and reading documents in English and writing reports. Practicing based on knowledge taught in class.		5%
A2	Projects			35%
A21	Project			35%
A3	Exams			45%
A31	Final exam	Closed book exam. Describe the understanding of different topics, analyze & program to solve problems.		45%

7. RESOURCES

Textbooks

- Jeff Hoffer, Ramesh Venkataraman, Heikki Topi, Modern Database System, 12th edition.
- Carlos Coronel, Steven Morris, Database System: Design, Implementation, and Management, 12th edition.
- Toby Teorey, Sam Lightstone, Tom Nadeau, Database Modeling and Design - Logical Design, 4th Edition.
- Sam Lightstone, Toby Teorey, Tom Nadeau, Physical Database Design, 4th Edition.

Others

- Data Design Tools: ER-Win, Power Designer.
- Programing Environments: C#, MS SQL Server.

8. GENERAL REGULATIONS & POLICIES

- All students are responsible for reading and following strictly the regulations and policies of the school and university.
- Students who are absent for more than 3 theory sessions are not allowed to take the exams.
- For any kind of cheating and plagiarism, students will be graded 0 for the course. The incident is then submitted to the school and university for further review.
- Students are encouraged to form study groups to discuss the topics. However, individual work must be done and submitted on your own.
- **Students will fail the course if they receive a score of 0 for the project (A2), or the score of 0 for the assignments (A1) or a final exam grade (A3) below 2, due to minimum grade requirements.**